

MASSCAN

ARPSCAN

Masscan & ARP-SCAN

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실습 & 분석

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MASSCAN

MASSCAN (Mass IP Port Scanner)

가장 빠른 인터넷 포트 스캐너.

6분안에 전체 인터넷을 스캔하여 초당 1,00만 패킷 전송 가능
nmap과 비슷한 결과값을 훨씬 빠르게 생성한다.

MASSCAN vs NMAP

MASSCAN

- 설계 목적이 속도
- 비동기식 -> 싸가지 없음.
- 속도가 빠르다.
- 열린 / 닫힌 포트 상태
- 대규모 네트워크, 인터넷 전체 스캔, 자산 인벤토리 등에 쓰임
- 포트 상태 확인에 최적화
- 동기식 : 요청을 보내고 응답 기다림 X
작업을 동시에
작업에 순서가 존재 X

NMAP

- 설계 목적이 정보의 깊이
- 동기식 -> 착한아이
- 비교적 느림
- 포트상태, 서비스/버전, OS종류, 취약점 스크립트 결과등
- 정밀분석, 보안 감사, 상세 정보 수집, 침투 테스트
- 정확도가 매우 높다.
- 동기식 : 요청을 보내고 응답을 기다린다.
순차적
작업에 순서가 존재

Option

대상 & 포트 지정 옵션

- --ports <ports> : 스캔할 포트 지정
- --includefile <FILE> : 스캔할 대상 IP 목록이 담긴 파일 지정
- --excludefile <FILE> : 스캔에서 제외할 IP 목록이 담긴 파일 지정

성능 & 네트워크 옵션

- --rate <RATE> : pps 지정
- -e <인터페이스> : 스캔 트래픽을 보낼 인터페이스 지정
- -S <IP> : 스캔에 사용할 출발지 IP 주소 지정
- -G <IP> : 스캔에서 사용할 출발지 게이트웨이 IP 지정
- --adapter-mac <MAC> : 스캔에 사용할 출발지 MAC 주소 지정
- --router-mac <MAC> : 라우터 / 게이트웨이의 MAC주소 지정

결과 및 출력 옵션

- -oX <FILE> : 결과를 XML 형식으로 저장 (NMAP 호환)
- -oG <FILE> : 결과를 grep 가능한 형식으로 저장한다. (NMAP의 -oG와 호환됨)
- -oL <FILE>: 결과를 리스트 형식으로 저장한다.
- -oJ <FILE>: 결과를 JSON 형식으로 저장한다.
- --output-format <포맷>: 출력형식을 명시적으로 지정한다.
- --out-filename<FILE>: 출력파일 이름을 명시적으로 지정한다.

ARP-SCAN

ARP-SCAN 이란

- ARP를 활용하여 동일 네트워크 내의 장치들을 빠르게 파악하는 데 사용됨.
- ARP요청을 브로드캐스트하여 응답하는 장치들의 IP주소와 MAC주소를 수집하고 매핑한다
- L2에서 작동하여 일부 방화벽을 우회할 수 있으며, MAC주소를 통해 장치의 제조사 정보도 함께 제공된다.

실습

환경설정

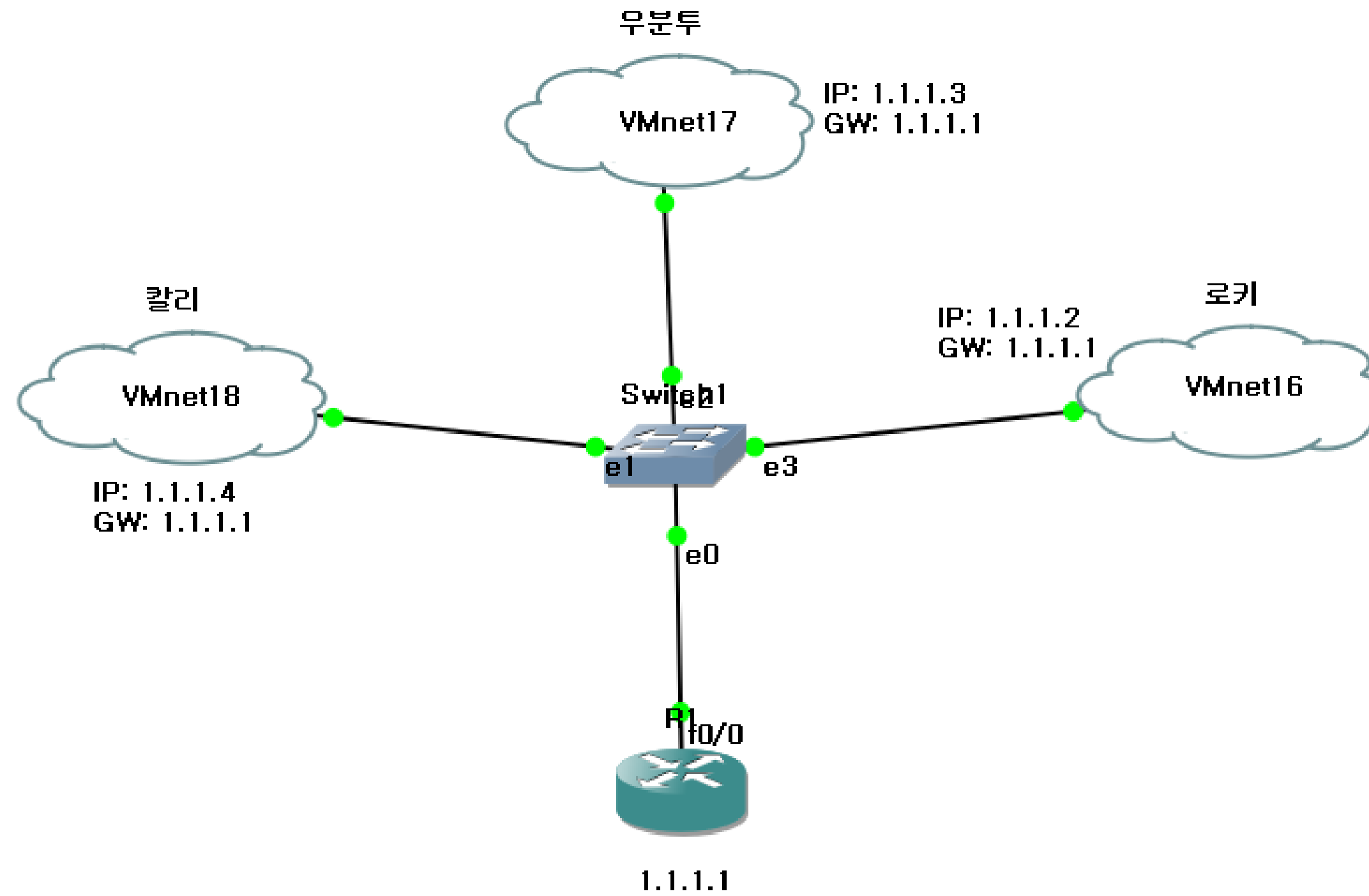
```
root@user:~# apt install tcpdump -y
패키지 tcpdump는 이미 최신 버전입니다 (4.99.5-2ubuntu2).
다음 패키지가 자동으로 설치되었지만 더 이상 필요하지 않습니다:
  galera-4          libdbi-perl      libterm-readkey-perl
  libconfig-inifiles-perl libpcres2-posix3 pv
'apt autoremove'를 이용하여 제거하십시오.
```

```
root@user:~# apt update -y && apt upgrade -y
받기:1 http://security.ubuntu.com/ubuntu questin
```

```
(root@kali)-[~]
# apt update -y && apt upgrade -y
받기:1 http://kali.download/kali kali-ro
받기:2 http://kali.download/kali kali-ro
```

```
[root@security ~]# dnf -y update && dnf -y upgrade
Last metadata expiration check: 0:33:52 ago on Wed Dec 17 00:42:48 2025.
Dependencies resolved.
Nothing to do.
Complete!
```


네트워크 맵 구성



환경설정

```
ens37: <BROADCAST,MULTICAST,UP,
h 1000
    link/ether 00:0c:29:99:3a:ef b
    altname enp2s5
    altname enx000c29993aef
    inet 1.1.1.3/24 brd 1.1.1.255
```

```
eth1: <BROADCAST,
default qlen 1000
    link/ether 00:0c
    inet 1.1.1.4/24
    valid lft for
```

```
ens224: <BROADCAST
    link/ether 00:0c
    altname enp19s0
    inet 1.1.1.2/24
    valid lft for
```

```
(kali㉿kali)-[~]
$ ping 1.1.1.2
PING 1.1.1.2 (1.1.1.2) 56(84) bytes of data.
64 bytes from 1.1.1.2: icmp_seq=1 ttl=64 time=3.61 ms
64 bytes from 1.1.1.2: icmp_seq=2 ttl=64 time=3.57 ms
^C
--- 1.1.1.2 ping statistics ---
2 packets transmitted, 2 received, 0% packet loss, time 1002ms
rtt min/avg/max/mdev = 3.565/3.588/3.611/0.023 ms

(kali㉿kali)-[~]
$ ping 1.1.1.3
PING 1.1.1.3 (1.1.1.3) 56(84) bytes of data.
64 bytes from 1.1.1.3: icmp_seq=1 ttl=64 time=3.40 ms
64 bytes from 1.1.1.3: icmp_seq=2 ttl=64 time=3.28 ms
```

```
[root@security ~]# ping 1.1.1.3
PING 1.1.1.3 (1.1.1.3) 56(84) bytes of data.
64 bytes from 1.1.1.3: icmp_seq=1 ttl=64 time=3.557 ms
64 bytes from 1.1.1.3: icmp_seq=2 ttl=64 time=3.557 ms
^C
--- 1.1.1.3 ping statistics ---
2 packets transmitted, 2 received, 0% packet loss, time 1002ms
rtt min/avg/max/mdev = 3.557/3.557/3.557/0.000 ms

[root@security ~]# ping 1.1.1.4
PING 1.1.1.4 (1.1.1.4) 56(84) bytes of data.
64 bytes from 1.1.1.4: icmp_seq=1 ttl=64 time=3.75 ms
64 bytes from 1.1.1.4: icmp_seq=2 ttl=64 time=2.92 ms
```

```
root@user:~# ping 1.1.1.2
PING 1.1.1.2 (1.1.1.2) 56(84) bytes of data.
64 bytes from 1.1.1.2: icmp_seq=1 ttl=64 time=3.42 ms
64 bytes from 1.1.1.2: icmp_seq=2 ttl=64 time=2.31 ms
^C
--- 1.1.1.2 ping statistics ---
2 packets transmitted, 2 received, 0% packet loss, time 1002ms
rtt min/avg/max/mdev = 2.310/2.864/3.419/0.554 ms

root@user:~# ping 1.1.1.4
PING 1.1.1.4 (1.1.1.4) 56(84) bytes of data.
64 bytes from 1.1.1.4: icmp_seq=1 ttl=64 time=3.75 ms
64 bytes from 1.1.1.4: icmp_seq=2 ttl=64 time=2.92 ms
64 bytes from 1.1.1.4: icmp_seq=3 ttl=64 time=3.00 ms
```

MASSCAN

```
(kali@kali) [~]  
$ masscan --help  
MASSCAN is a fast port scanner. The primary input parameters are the  
IP addresses/ranges you want to scan, and the port numbers. An example  
is the following, which scans the 10.x.x.x network for web servers:  
masscan 10.0.0.0/8 -p80  
The program auto-detects network interface/adapter settings. If this  
fails, you'll have to set these manually. The following is an  
example of all the parameters that are needed:  
--adapter-ip 192.168.10.123  
--adapter-mac 00-11-22-33-44-55  
--router-mac 66-55-44-33-22-11
```

MASSCAN

```
(root@kali)-[~]  
# masscan 1.1.1.0/24 -p22,80 --rate 100  
Starting masscan 1.3.2 (http://bit.ly/14GZzcI)  
Initiating CWM Stealth Scan
```

```
(root@kali)-[~]  
# masscan 1.1.1.0/24 -p22,80 --rate 10000  
Starting masscan 1.3.2 (http://bit.ly/14GZzcI)
```

```
(root@kali)-[~]  
# masscan 1.1.1.2 -p1-65535 --rate 10000
```

칼리에서 터미널을 루트권한으로 실행한다.

위 2개의 명령어를 입력하고 차이점을 찾아보자

그리고 둘 명령어의 출력값으로 이상한것을 찾아보자

둘다 한 후에 맨 아래 코드를 실행하여 보자

MASSCAN - Wireshark

1	0.000000	172.16.28.140	1.1.1.2	TCP	60	40788 → 2095 [RST] Seq=1 Win=0 Len=0
2	0.434077	172.16.28.140	1.1.1.2	TCP	60	40788 → 8443 [RST] Seq=1 Win=0 Len=0
3	0.698401	172.16.28.140	1.1.1.2	TCP	60	40788 → 2082 [RST] Seq=1 Win=0 Len=0
66	6.088548	172.16.28.140	1.1.1.2	TCP	60	40788 → 8880 [RST] Seq=1 Win=0 Len=0
67	7.119124	172.16.28.140	1.1.1.2	TCP	60	40788 → 8880 [RST] Seq=1 Win=0 Len=0
68	8.598229	172.16.28.140	1.1.1.2	TCP	60	40788 → 443 [RST] Seq=1 Win=0 Len=0
69	9.491070	172.16.28.140	1.1.1.2	TCP	60	40788 → 853 [RST] Seq=1 Win=0 Len=0
70	9.646475	172.16.28.140	1.1.1.2	TCP	60	40788 → 2096 [RST] Seq=1 Win=0 Len=0
71	9.752447	172.16.28.140	1.1.1.2	TCP	60	40788 → 8080 [RST] Seq=1 Win=0 Len=0
72	10.058531	172.16.28.140	1.1.1.2	TCP	60	40788 → 2087 [RST] Seq=1 Win=0 Len=0
73	10.177509	172.16.28.140	1.1.1.2	TCP	60	40788 → 80 [RST] Seq=1 Win=0 Len=0
74	10.784519	172.16.28.140	1.1.1.2	TCP	60	40788 → 2083 [RST] Seq=1 Win=0 Len=0
77	11.131167	172.16.28.140	1.1.1.2	TCP	60	40788 → 53 [RST] Seq=1 Win=0 Len=0
78	11.219114	172.16.28.140	1.1.1.2	Frame 73: 60 bytes on wire (480 bits), 60 bytes captured (480 bits) on interface Ethernet II, Src: VMware_07:b3:ca (00:0c:29:07:b3:ca), Dst: VMware_a9:09:10 (00:0c:29:a9:09:10) Internet Protocol Version 4, Src: 172.16.28.140, Dst: 1.1.1.2 Transmission Control Protocol, Src Port: 40788, Dst Port: 80, Seq: 1, Len: 0		
79	11.228637	172.16.28.140	1.1.1.2			

잡 패킷이 뜨니 TCP만 필터링 걸어서 패킷들의 하나하나 포트들을 확인해준다.

TCPDUMP

```
480479, win 1024, length 0
01:10:14.553890 IP 172.16.28.140.41408 > family.cloudflare-dns.com.9377: Flags [S], seq 19041
4765, win 1024, length 0
01:10:14.553890 IP 172.16.28.140.41408 > family.cloudflare-dns.com.29307: Flags [S], seq 5768
9685, win 1024, length 0
01:10:14.553890 IP 172.16.28.140.41408 > family.cloudflare-dns.com.7849: Flags [S], seq 11148
25866, win 1024, length 0
01:10:14.553926 IP 172.16.28.140.41408 > family.cloudflare-dns.com.30548: Flags [S], seq 3550
089163, win 1024, length 0
01:10:14.553931 IP 172.16.28.140.41408 > family.cloudflare-dns.com.32511: Flags [S], seq 3124
631354, win 1024, length 0
01:10:14.553957 IP 172.16.28.140.41408 > family.cloudflare-dns.com.20901: Flags [S], seq 2119
724366, win 1024, length 0
01:10:14.553998 IP 172.16.28.140.41408 > family.cloudflare-dns.com.52068: Flags [S], seq 3892
243139, win 1024, length 0
01:10:14.553998 IP 172.16.28.140.41408 > family.cloudflare-dns.com.16650: Flags [S], seq 1263
092096, win 1024, length 0
01:10:14.556540 IP 172.16.28.140.41408 > family.cloudflare-dns.com.34196: Flags [S], seq 2309
271919, win 1024, length 0
```

apt install tcpdump -> tcpdump --version

ARPSCAN

```
valid_lft  
3: eth1: <BROADCAST  
roup default qlen  
link/ether 00  
inet 1.1.1.4/
```

```
(root@kali) ~#  
# arp-scan --interface=eth1 1.1.1.0/24  
Interface: eth1, type: EN10MB, MAC: 00:0c:29:07:b3:ca, IPv4: 1.1.1.4  
Starting arp-scan 1.10.0 with 256 hosts (https://github.com/royhills/arp-scan)  
  
1.1.1.2 00:0c:29:a9:09:10 VMware, Inc.  
1.1.1.3 00:0c:29:99:3a:ef VMware, Inc.  
1.1.1.1 ca:01:3d:48:00:00 (Unknown: locally administered)  
  
0 packets received by filter, 0 packets dropped by kernel  
Ending arp-scan 1.10.0: 256 hosts scanned in 2.004 seconds (127.74 hosts/sec)  
3 responded
```

인터페이스 확인후 오른쪽의 코드와 같이 입력을 해주면 해당 IP주소와 MAC주소를 매핑하여 출력해준다

ARPSCAN-wireshark

475	0.955998	VMware_07:b3:ca	Broadcast	ARP	60	Who has 1.1.1.220? Tell 1.1.1.4
476	0.959308	VMware_07:b3:ca	Broadcast	ARP	60	Who has 1.1.1.221? Tell 1.1.1.4
477	0.962816	VMware_07:b3:ca	Broadcast	ARP	60	Who has 1.1.1.222? Tell 1.1.1.4
478	0.965301	VMware_07:b3:ca	Broadcast	ARP	60	Who has 1.1.1.223? Tell 1.1.1.4
479	0.966297	VMware_07:b3:ca	Broadcast	ARP	60	Who has 1.1.1.224? Tell 1.1.1.4
480	0.966788	VMware_07:b3:ca	Broadcast	ARP	60	Who has 1.1.1.225? Tell 1.1.1.4
481	0.968287	VMware_07:b3:ca	Broadcast	ARP	60	Who has 1.1.1.226? Tell 1.1.1.4
482	0.971287	VMware_07:b3:ca	Broadcast	ARP	60	Who has 1.1.1.227? Tell 1.1.1.4
483	0.974829	VMware_07:b3:ca	Broadcast	ARP	60	Who has 1.1.1.228? Tell 1.1.1.4
484	0.977402	VMware_07:b3:ca	Broadcast	ARP	60	Who has 1.1.1.229? Tell 1.1.1.4
485	0.978342	VMware_07:b3:ca	Broadcast	ARP	60	Who has 1.1.1.230? Tell 1.1.1.4
486	0.978801	VMware_07:b3:ca	Broadcast	ARP	60	Who has 1.1.1.231? Tell 1.1.1.4
487	0.980312	VMware_07:b3:ca	Broadcast	ARP	60	Who has 1.1.1.232? Tell 1.1.1.4
488	0.983313	VMware_07:b3:ca	Broadcast	ARP	60	Who has 1.1.1.233? Tell 1.1.1.4
489	0.986814	VMware_07:b3:ca	Broadcast	ARP	60	Who has 1.1.1.234? Tell 1.1.1.4
490	0.989356	VMware_07:b3:ca	Broadcast	ARP	60	Who has 1.1.1.235? Tell 1.1.1.4
491	0.990368	VMware_07:b3:ca	Broadcast	ARP	60	Who has 1.1.1.236? Tell 1.1.1.4
492	0.990490	VMware_07:b3:ca	Broadcast	ARP	60	Who has 1.1.1.237? Tell 1.1.1.4
493	0.992363	VMware_07:b3:ca	Broadcast	ARP	60	Who has 1.1.1.238? Tell 1.1.1.4
494	0.995340	VMware_07:b3:ca	Broadcast	ARP	60	Who has 1.1.1.239? Tell 1.1.1.4
495	0.998040	VMware_07:b3:ca	Broadcast	ARP	60	Who has 1.1.1.240? Tell 1.1.1.4

ARPSCAN-TCPdump(U)

```
root@user:/home/han1# tcpdump -i ens37 arp
tcpdump: verbose output suppressed, use -v[v]... for full protocol decode
listening on ens37, link-type EN10MB (Ethernet), snapshot length 262144 bytes
01:41:49.514136 ARP, Request who-has 1.1.1.0 tell 1.1.1.4, length 46
01:41:49.516239 ARP, Request who-has one.one.one.one tell 1.1.1.4, length 46
01:41:49.518632 ARP, Request who-has security.cloudflare-dns.com tell 1.1.1.4, length 46
01:41:49.520191 ARP, Request who-has family.cloudflare-dns.com tell 1.1.1.4, length 46
01:41:49.520198 ARP, Reply family.cloudflare-dns.com is-at 00:0c:29:99:3a:ef (oui) length 28
01:41:49.522184 ARP, Request who-has 1.1.1.4 tell 1.1.1.4, length 46
01:41:49.523764 ARP, Request who-has 1.1.1.5 tell 1.1.1.4, length 46
01:41:49.525729 ARP, Request who-has 1.1.1.6 tell 1.1.1.4, length 46
01:41:49.527005 ARP, Request who-has 1.1.1.7 tell 1.1.1.4, length 46
```

ARPSCAN-s IP, MAC주소 위조

```
)  
1.1.1.2 00:0c:29:a9:09:10      VMware, Inc.  
1.1.1.3 00:0c:29:99:3a:ef      VMware, Inc.  
1.1.1.1 ca:01:3d:48:00:00      (Unknown: locally administered)  
  
3 packets received by filter, 0 packets dropped by kernel  
Ending arp-scan 1.10.0: 256 hosts scanned in 2.021 seconds (126.67  
. 3 responded  
  
└─(root@kali)-[~]  
└─# arp-scan --arp-spa=1.1.1.2 --arp-sha=00:0c:29:a9:09:10 1.1.1.3
```

ARPSCAN-s IP, MAC주소 위조

VMware_a9:09:10	ca:01:3d:48:00:00	ARP	42 Who has 1.1.1.1? Tell 1.1.1.2
ca:01:3d:48:00:00	VMware_a9:09:10	ARP	60 1.1.1.1 is at ca:01:3d:48:00:00

그래서 이걸로 뭐할건가요~? ^^라면

ARP-SCAN

- 내부에 들어온 직후, 살아있는 호스트들을 식별
- 살아있는 IP
- MAC주소
- 벤더 정보 (서버/PC/네트워크 장비 구분)

MASSCAN

- ARP SCAN 이후 공격 가능한 서비스 포트를 뽑아낸다
- 열린포트
- 서비스 후보 - SSH, HTTP etc
- 포트 분포 -> 이 서버는 뭐하는 서버일까? 맞출수 있음

감사합니다.